

RISHIKESH BHYRI

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EDUCATION

University at Buffalo, The State University of New York

Buffalo, NY

Master of Science in Computer Science – **CGPA: 3.95/4.0**

Aug 2024 – Jun 2026

- Thesis: Structured Spatial Reasoning for Robust and Transparent Object Counting
- Advisor: Prof. Junsong Yuan
- Coursework: Machine Learning, Parallel and Distributed Processing, Reinforcement Learning, Deep Learning

Vellore Institute of Technology

Chennai, India

Bachelor of Technology in Electronics and Communication Engineering – **CGPA: 9.48/10.0**

Jul 2017 – Jun 2021

- Capstone: Logo Detection and Analysis in Images.
- Advisor: Prof. Sathiya Narayanan. S

PUBLICATIONS AND PATENTS

- **Rishikesh Bhyri** et al., “Chain-of-Look Spatial Reasoning for Dense Surgical Instrument Counting” IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2026. **(Accepted)** (Paper)
- Multi-Agent Computer Vision system and methods for automated object counting, classification and tracking. USPTO Application No. 63/851,812. Status: **Patent Pending**
- Philip J Seger, Shinil Shah, ..., **Rishikesh Bhyri**, et al., “Computer Vision Augmented Indocyanine Green-Near Infrared Fluorescence (ICG-NIRF) Signal-to-Background Ratio (SBR) for Identification of Extrahepatic Biliary Structures in Laparoscopic Cholecystectomy,” Surgical Endoscopy, 2026. **(Accepted)** (Oral Presentation at SAGES 2026)

GRANTS, HONORS & AWARDS

- **NSF I-Corps Regional Grant & Travel Award, Cornell University & NSF** (Oct 2026): Selected for the I-Corps MedTech Regional Course; awarded \$5,000 in grant funding to conduct on-site clinical customer discovery and commercialization validation at The MedTech Conference 2026 (Boston, MA).
- **CSE Faculty Choice Graduate Award, University at Buffalo** (May 2026): Awarded by the department committee; invited graduate speaker for the CSE graduation dinner.
- **Best AI/ML Project, UBHacking’24** (Nov 2024): Won the ‘Best AI/ML Project’ award for developing an AI-powered, context-aware bookmark manager extension for Chrome.
- **Top Performance Rating, Citi**: Awarded a top rating (1 out of 4) for innovation in developing automated reporting tools and dashboards.
- **Academic Meritorious Award**: Achieved 4th overall department rank in B.Tech Electronics and Communication Engineering (2017–2021 batch).

RESEARCH EXPERIENCE

Graduate Research Assistant

Buffalo, NY

University at Buffalo (Advisor: Prof. Junsong Yuan, Dr. Peter C W Kim)

Oct 2024 – Present

- **High-Density Object Counting**: Engineered a novel framework extending CountGD (GroundingDINO, Swin-B, BERT) by integrating class-specific learnable tokens and a domain-specific loss function to enhance spatial reasoning in dense clusters. Curated the SurgCount-HD dataset (1,464 clinical images), achieving a state-of-the-art Mean Absolute Error (MAE) of 0.88.
- **Density-Aware Counting**: Building upon the initial CountGD counting framework, fine-tuned SAM3 for text-prompted, class-agnostic instance segmentation across both density regimes. Scaled dataset curation via a continuous data flywheel annotation strategy, achieving 0.774 AP / 0.178 MAE on low-density data (2,265 images) and 0.765 AP / 0.70 MAE on high-density data (800 images).
- **Calibrated Uncertainty & Self-Refinement**: Built a lightweight MLP risk-estimation model over frozen detector features to predict mask failures, achieving high calibration (ECE 0.065–0.088). Integrated a feedback loop that automatically re-processes uncertain image regions at higher resolutions before confirming the count.

- **Human-in-the-Loop (HITL) & Operating Room (OR) Deployment:** Designed an on-device UI that routes calibrated risk scores to cluster unverified edge cases into localized crops for human review. Deployed the AbacuSx mobile app in live surgical trials ($n = 15$ surgeries), maintaining 99.91% accuracy while reducing counting time to 6–8 seconds.
- **3D Digital Twin Generation (Ongoing):** Developing a custom SAM3 and NeuS2 pipeline to construct a 3D model catalog of surgical instruments, enabling synthetic dataset scaling and digital-twin-based verification for future architectures.

Computer Vision Intern

Remote

PlaEye LLC (Portland, OR) (Supervisor: Mr. Venkat Yellepeddy)

Jan 2021 – Jun 2021

- **Logo Detection and Analysis in Images:** Developed a **mobile-based brand analytics system** for **logo detection and classification** using the LogoDet-3k dataset, optimized for arm64-v8a devices with ArmNN. Addressed challenges in dataset imbalance and diverse image features by implementing scaled loss functions and class augmentations. ([Link](#))
- **Automatic Calibration of Tennis Court Images:** Created a novel algorithm for automated tennis **court calibration using Homography and Cross ratios** to overcome **single-view calibration limitations** arising from elusive feature points in occluded views. ([Link](#))

PROFESSIONAL EXPERIENCE

Machine Learning Intern

San Jose, CA

Mercedes-Benz R&D North America

May 2025 – Aug 2025

- Optimized inference by exploring **CUDA Graphs** with **Nsight Systems**, trimming CPU launch overhead and **cutting latency 20%**.
- Drove **4× faster inference** via INT8 **implicit and explicit quantization**.
- Built a C++/CUDA utility to inspect TensorRT engine weights and automate network-pruning sweeps.
- Implemented **unified-memory zero-copy** paths and custom GPU pools, **lowering peak GPU usage by 18%** and eliminating two memcpy calls per frame.
- Proved bit-exact **determinism** at runtime & layer level; stress-tested execution contexts to meet **ASIL-D safety**.
- Designated as **Primary Inventor** for **2 patents** (currently in filing stage).

Software Engineer

Chennai, India

Citi

Jul 2021 – Jul 2024

- Co-owned and led the development of a web-based end-to-end tool initiative for data creation and conditioning, reducing the time and effort required for data handling. Engineered the tool using ReactJS and NodeJS.
- Developed API layers in .NET using minimal-API, exposing Selenium-C# automation functionalities and enabling remote headless execution which **decreased data creation time by 5x**, reduced cross-team dependency and **turnaround time by 80%**.

RECENT PROJECTS

Medical Video Diffusion Reasoning (MVDR) | [Submitted to ECCV 2026](#) | [Paper](#)

Jun 2026

- Developed a novel framework that formulates step-by-step visual reasoning in clinical videos as a continuous diffusion process within a latent reasoning space.
- Engineered a conditional denoiser using a 1D U-Net with Adaptive Layer Normalization (AdaLN) to progressively refine noisy initial frame embeddings into interpretable temporal reasoning trajectories.
- Evaluated the architecture on the OphNet and HyperKvasir medical datasets, achieving performance comparable to state-of-the-art vision models while generating transparent spatio-temporal heatmaps to explain decision-making.

Masked-ResShift: Pixel-Level Residual Shift for Image Inpainting | [Paper](#)

Jan 2025 – May 2025

- Optimized diffusion-based image inpainting by implementing novel strategies, including Masked-Variance Diffusion (MVD) and Exact Masked-Markov Diffusion (EMMD), achieving faster inference times while maintaining high image quality.

Traffic light automation using Reinforcement Learning | [Link](#)

Jan 2025 – May 2025

- Designed a dynamic traffic control system using SUMO-RL to minimize wait times for emergency vehicles.
- Trained and benchmarked agents using **SARSA, DQN, Double DQN, and A2C** algorithms, demonstrating superior performance over static baselines in reducing emergency transit delays.

- MindVault: AI-Powered Bookmark Organizer** | *UBHacking'24 Best AI/ML Project*

Nov 2024
- Chrome extension using BeautifulSoup, NLTK, Sentence Transformers, and DBSCAN for context-aware bookmark clustering.
- Open-Source Contribution** | *Segment Anything Model 3 (SAM 3)*

Nov 2025
- Assisted users by answering queries regarding fine-tuning bugs and model configuration.
 - Provided debugging support that helped close 3 GitHub issues: [#200](#), [#260](#), and [#286](#).

ACADEMIC SERVICE

- Peer Reviewer**

Machine Vision and Applications (MVA) Journal, Springer

Feb 2026 – Present
- Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP) 2026*

Sept 2026 – Present
- Graduate Student Assistant – Grader**

University at Buffalo (Supervisor: Prof. Asif Imran)

Buffalo, NY
Jan 2025 – May 2025
- Evaluated programming assignments, quizzes, and coursework for **CSE574: Introduction to Machine Learning**.
- Teaching Assistant**

Vellore Institute of Technology (Supervisor: Prof. Vetrivelan P)

Chennai, India
Jan 2020 – May 2020
- Evaluated programming assignments and quizzes for **ECE2005: Probability Theory and Random Process**.

TECHNICAL SKILLS

- Languages:** Python, C++, OpenMP, MPI, CUDA
- Tools:** PyTorch, TensorRT, ArmNN, ONNX, OpenCV, Scikit-Learn, Git, CMake, GCP, Slurm, Docker, LaTeX
- Concepts:** Deep Learning, Computer Vision, Vision Transformers, Diffusion Models, Object Detection, Data Structures, Linear Algebra, Object Oriented Programming, VLM, PEFT